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Electronic device for performing medium synchronization of a link and method for the same

Abstract

In an electronic device and an operation method thereof according to various embodiments, the electronic device may include: at least one antenna; a communication circuit electrically connected to the antenna and configured to transmit and receive data through a first link and/or a second link established between an external electronic device and the electronic device; and a processor operably connected to the communication circuit, wherein the processor may be configured to: receive, in a state of transmitting data to the external electronic device through the first link, medium synchronization information of the second link through the first link; and perform synchronization of the second link based on the medium synchronization information.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/KR2022/008051 designating the United States, filed on Jun. 8, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0094120, filed on Jul. 19, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND
Field
(1) The disclosure relates to an electronic device and operation method thereof, and to a technique for performing medium synchronization of a link established between an external electronic device and the electronic device.
Description of Related Art
(2) With the popularization of various electronic devices, speed improvement for wireless communication that can be used by various electronic devices has been achieved. Among wireless communications supported by recent electronic devices, IEEE 802.11 WLAN (or Wi-Fi) is a standard for implementing high-speed wireless connections between various electronic devices. The first implemented Wi-Fi could support a transmission rate of up to 1 to 9 Mbps, but Wi-Fi 6 technology (or IEEE 802.11ax) can support a transmission rate of up to about 10 Gbps.
(3) Electronic devices may support various services utilizing relatively large amounts of data (e.g., a UHD video streaming service, an augmented reality (AR) service, a virtual reality (VR) service, a mixed reality (MR) service) through wireless communication supporting high transmission rates, and may support various other services.
(4) It has been proposed that the IEEE 802.11 WLAN standard introduce a technology supporting multi-link operation (MLO) in order to improve data transmission and reception speed and reduce latency. An electronic device supporting multi-link operation may transmit or receive data through a plurality of links, and thus it is expected to achieve a relatively high transmission speed and a low delay time.
(5) In a wireless LAN system, an electronic device may use a method of carrier sense multiple access with collision avoidance (CSMA/CA) to prevent a collision caused by transmitting data simultaneously through the same link as other electronic devices. The CSMA/CA method is a method of performing data transmission when a specific link is in idle state, and the electronic device supporting CSMA/CA may check whether another electronic device transmits data through a specific link and transmit data if the other electronic device does not transmit data through the specific link. An electronic device supporting multi-link operation may transmit data by using the CSMA/CA method for each of plural links.
(6) Considering that interference may occur between links due to the limitation of the mounting space of an electronic device, the IEEE 802.11 WLAN standard considers support of a non-STA mode or an enhanced multi-link single radio (EMLSR) mode in which, when data is being

transmitted to an external electronic device through one link, data is not received through another link.
(7) To transmit data through a specific link, the electronic device may determine whether the specific link is in idle state. In response to determining that the specific link is in idle state, the electronic device may transmit data through the specific link. To determine whether a specific link is in idle state, the electronic device may perform medium synchronization of the specific link. Medium synchronization may refer to, for example, an operation of tracking the status of a specific link in order to determine whether the specific link is in an occupied state or how long the occupied state will be maintained. Medium synchronization may be performed based on data received through a link that is a target of the medium synchronization.
(8) However, the electronic device operating in non-STR mode or EMLSR mode may be unable to receive data through a specific link while transmitting data through another link, in which case it may fail to receive information for medium synchronization of the specific link.
(9) When the medium synchronization of a specific link is released, the electronic device may wait until data is received through the specific link or may be unable to perform data transmission until a specified time (e.g., time-out), which may increase the latency of data transmission.
SUMMARY
(10) An electronic device according to various example embodiments of the disclosure may include: at least one antenna; a communication circuit electrically connected to the antenna and configured to transmit and receive data through a first link and a second link established between an external electronic device and the electronic device; and a processor operably connected to the communication circuit, wherein the processor may be configured to: receive, in a state of transmitting data to the external electronic device through the first link, medium synchronization information of the second link through the first link; and perform synchronization of the second link based on the medium synchronization information.
(11) An electronic device according to various example embodiments of the disclosure may include: at least one antenna; a communication circuit electrically connected to the antenna and configured to transmit and receive data through a first link and a second link established between an external electronic device and the electronic device; and a processor operably connected to the communication circuit, wherein the processor may be configured to: generate, in a state of receiving data from the external electronic device through the first link, medium synchronization information of the second link; and control, upon completion of reception of the data, the communication circuit to transmit the medium synchronization information of the second link to the external electronic device through the first link, and wherein the medium synchronization information may include information about a timer for medium synchronization of the second link.
(12) A method of operating an electronic device according to various example embodiments of the disclosure may include: receiving, in a state of transmitting data through a first link established between an external electronic device and the electronic device, medium synchronization information of a second link; and performing synchronization of the second link based on the medium synchronization information.
(13) The electronic device and operation method thereof according to various example embodiments of the disclosure can receive medium synchronization information of a second link through a first link while transmitting data through the first link, and perform or maintain medium synchronization of the second link based on the medium synchronization information of the second link. Accordingly, even if the electronic device operates in non-STR mode or EMLSR mode, it can perform data transmission rapidly through the second link, so that data transmission speed can be improved and latency can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1** is a block diagram illustrating an example electronic device in a network environment according to various embodiments;

(3) FIG. **2** is a block diagram illustrating an example configuration of programs according to various embodiments;

(4) FIG. **3** is a diagram illustrating an example in which an electronic device and an access point (AP) operate in multi-link operation (MLO) according to various embodiments;

(5) FIG. **4**A is a diagram illustrating an example in which electronic devices perform medium synchronization of a link according to various embodiments;

(6) FIG. **4**B is a diagram illustrating an example in which the electronic device operates in non-simultaneous transmission and reception (non-STR) mode according to various embodiments;

(7) FIG. **4**C is a diagram illustrating an example in which the electronic device operates in enhanced multi-link with single radio (EMLSR) mode according to various embodiments;

(8) FIG. **5** is a block diagram illustrating an example configuration of an electronic device according to various embodiments;

(9) FIG. **6** is a block diagram illustrating an example configuration of an AP according to various embodiments;

(10) FIG. **7** is a diagram illustrating an example in which the electronic device performs medium synchronization of a second link by using medium synchronization information received through a first link according to various embodiments;

(11) FIG. **8** is a diagram illustrating an example structure of a frame including information for the electronic device to perform medium synchronization of a second link according to various embodiments;

(12) FIG. **9** is a flowchart illustrating an example method of operating an electronic device according to various embodiments; and

(13) FIG. **10** is a flowchart illustrating an example method of operating an electronic device according to various embodiments.

DETAILED DESCRIPTION

(14) FIG. **1** is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connection terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connection terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In various embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(15) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101**

coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.
(16) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
(17) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
(18) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
(19) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
(20) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of, the speaker.
(21) The display module **160** may visually provide information to the outside (e.g., a user) of the

electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(22) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(23) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(24) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(25) A connection terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(26) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(27) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(28) The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(29) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(30) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module).

A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(31) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(32) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(33) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(34) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(35) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.
(36) FIG. **2** is a block diagram **200** illustrating the program **140** according to various embodiments. According to an embodiment, the program **140** may include an operating system (OS) **142** to control one or more resources of the electronic device **101**, middleware **144**, or an application **146** executable in the OS **142**. The OS **142** may include, for example, Android™, iOS™, Windows™, Symbian™, Tizen™, or Bath™. At least part of the program **140**, for example, may be pre-loaded on the electronic device **101** during manufacture, or may be downloaded from or updated by an external electronic device (e.g., the electronic device **102** or **104**, or the server **108**) during use by a user.
(37) The OS **142** may control management (e.g., allocating or deallocation) of one or more system resources (e.g., process, memory, or power source) of the electronic device **101**. The OS **142**, additionally or alternatively, may include one or more driver programs to drive other hardware devices of the electronic device **101**, for example, the input module **150**, the sound output module **155**, the display module **160**, the audio module **170**, the sensor module **176**, the interface **177**, the haptic module **179**, the camera module **180**, the power management module **188**, the battery **189**, the communication module **190**, the subscriber identification module **196**, or the antenna module **197**.
(38) The middleware **144** may provide various functions to the application **146** such that a function or information provided from one or more resources of the electronic device **101** may be used by the application **146**. The middleware **144** may include, for example, an application manager **201**, a window manager **203**, a multimedia manager **205**, a resource manager **207**, a power manager **209**, a database manager **211**, a package manager **213**, a connectivity manager **215**, a notification manager **217**, a location manager **219**, a graphic manager **221**, a security manager **223**, a telephony (call) manager **225**, or a voice recognition manager **227**.
(39) The application manager **201**, for example, may manage the life cycle of the application **146**. The window manager **203**, for example, may manage one or more graphical user interface (GUI) resources that are used on a screen. The multimedia manager **205**, for example, may identify one or more formats to be used to play media files, and may encode or decode a corresponding one of the media files using a codec appropriate for a corresponding format selected from the one or more

formats. The resource manager **207**, for example, may manage the source code of the application **146** or a memory space of the memory **130**. The power manager **209**, for example, may manage the capacity, temperature, or power of the battery **189**, and determine or provide related information to be used for the operation of the electronic device **101** based at least in part on corresponding information of the capacity, temperature, or power of the battery **189**. According to an embodiment, the power manager **209** may interwork with a basic input/output system (BIOS) (not shown) of the electronic device **101**.

(40) The database manager **211**, for example, may generate, search, or change a database to be used by the application **146**. The package manager **213**, for example, may manage installation or update of an application that is distributed in the form of a package file. The connectivity manager **215**, for example, may manage a wireless connection or a direct connection between the electronic device **101** and the external electronic device. The notification manager **217**, for example, may provide a function to notify a user of an occurrence of a specified event (e.g., an incoming call, message, or alert). The location manager **219**, for example, may manage locational information on the electronic device **101**. The graphic manager **221**, for example, may manage one or more graphic effects to be offered to a user or a user interface related to the one or more graphic effects.

(41) The security manager **223**, for example, may provide system security or user authentication. The telephony (call) manager **225**, for example, may manage a voice call function or a video call function provided by the electronic device **101**. The voice recognition manager **227**, for example, may transmit a user's voice data to the server **108**, and receive, from the server **108**, a command corresponding to a function to be executed on the electronic device **101** based at least in part on the voice data, or text data converted based at least in part on the voice data. According to an embodiment, the middleware **244** may dynamically delete some existing components or add new components. According to an embodiment, at least part of the middleware **144** may be included as part of the OS **142** or may be implemented as another software separate from the OS **142**.

(42) The application **146** may include, for example, a home **251**, dialer **253**, short message service (SMS)/multimedia messaging service (MMS) **255**, instant message (IM) **257**, browser **259**, camera **261**, alarm **263**, contact **265**, voice recognition **267**, email **269**, calendar **271**, media player **273**, album **275**, watch **277**, health **279** (e.g., for measuring the degree of workout or biometric information, such as blood sugar), or environmental information **281** (e.g., for measuring air pressure, humidity, or temperature information) application. According to an embodiment, the application **146** may further include an information exchanging application (not shown) that is capable of supporting information exchange between the electronic device **101** and the external electronic device. The information exchange application, for example, may include a notification relay application adapted to transfer designated information (e.g., a call, message, or alert) to the external electronic device or a device management application adapted to manage the external electronic device. The notification relay application may transfer notification information corresponding to an occurrence of a specified event (e.g., receipt of an email) at another application (e.g., the email application **269**) of the electronic device **101** to the external electronic device. Additionally or alternatively, the notification relay application may receive notification information from the external electronic device and provide the notification information to a user of the electronic device **101**.

(43) The device management application may control the power (e.g., turn-on or turn-off) or the function (e.g., adjustment of brightness, resolution, or focus) of the external electronic device or some component thereof (e.g., a display module or a camera module of the external electronic device). The device management application, additionally or alternatively, may support installation, delete, or update of an application running on the external electronic device.

(44) FIG. **3** is a diagram illustrating an embodiment in which an electronic device and an external electronic device (e.g., access point (AP)) operate in multi-link operation (MLO) according to various embodiments of the disclosure.

(45) With reference to FIG. **3**, the wireless LAN system **300** may include an electronic device **310** and/or an external electronic device **320**. According to an embodiment, the electronic device **310** may perform wireless communication with the external electronic device **320** through short-range wireless communication. Wireless communication may refer to various communication methods that both the electronic device **310** and/or the external electronic device **320** can support. For example, wireless communication may be Wi-Fi. The external electronic device **320** may serve as a base station that provides wireless communication to at least one electronic device **310** located inside the communication radius of the wireless LAN system **300**. For example, the external electronic device **320** may include an access point (AP) of IEEE 802.11. The electronic device **310** may include a station (STA) of IEEE 802.11.
(46) According to various embodiments of the disclosure, the electronic device **310** and/or the external electronic device **320** may support multi-link operation (MLO). The multi-link operation may be an operation mode in which data is transmitted or received through plural links (e.g., first link **331** and second link **332**). The multi-link operation is an operation mode to be introduced in IEEE 802.11be, and may be an operation mode in which data is transmitted or received through plural links based on plural bands or channels.
(47) According to various embodiments of the disclosure, the electronic device **310** may include a plurality of communication circuits (e.g., first communication circuit **311** and/or second communication circuit **312**) to support multi-link operation. The first communication circuit **311** may transmit data to the external electronic device **320** through a first link **331** or receive data transmitted by the external electronic device **320** through the first link **331**. The first communication circuit **311** may output or receive a signal of a frequency band corresponding to the first link **331** through a first antenna **313**. The second communication circuit **312** may transmit data to the external electronic device **320** through a second link **332** or receive data transmitted by the external electronic device **320** through the second link **332**. The second communication circuit **312** may output or receive a signal of a frequency band corresponding to the second link **332** through a second antenna **314**.
(48) According to various embodiments of the disclosure, the external electronic device **320** may include a plurality of communication circuits (e.g., third communication circuit **321** and/or fourth communication circuit **322**) to support multi-link operation. The third communication circuit **321** may transmit data to the electronic device **310** through the first link **331** or receive data transmitted by the electronic device **310** through the first link **331**. The third communication circuit **321** may output or receive a signal of a frequency band corresponding to the first link **331** through a third antenna **323**. The fourth communication circuit **322** may transmit data to the electronic device **310** through the second link **332** or receive data transmitted by the electronic device **310** through the second link **332**. The fourth communication circuit **322** may output or receive a signal of a frequency band corresponding to the second link **332** through a fourth antenna **324**.
(49) According to various embodiments of the disclosure, the frequency band of the first link **331** and the frequency band of the second link **333** may be different from each other. For example, the frequency band of the first link **331** may be 2.5 GHz, and the frequency band of the second link **332** may be 5 GHz or 6 GHz.
(50) According to various embodiments of the disclosure, a different electronic device other than the electronic device **310** may also use the first link **331** and the second link **332**. To prevent a situation in which the electronic device **310** and another electronic device transmit or receive data through the same link at the same time, the electronic device **310** may support a carrier sense multiple access with collision avoidance (CSMA/CA) method. The CSMA/CA method may, for example, refer to a scheme for performing data transmission when a specific link is in idle state. The electronic device **310** supporting CSMA/CA may check whether another electronic device transmits data through a specific link, and, upon sensing data transmission, may wait without transmitting data through the specific link. Upon confirming that another electronic device does not

transmit data through the specific link, the electronic device **310** supporting CSMA/CA may transmit data through the specific link according to a specified scheme (e.g., activating a timer and transmitting data when the timer expires). In this way, the electronic device **310** may perform data transmission and/or reception through a specific link without colliding with another electronic device.

(51) According to various embodiments of the disclosure, the first link **331** and/or the second link **332** supported by the multi-link operation may independently support CSMA/CA.

(52) FIG. **4**A is a diagram illustrating an example in which electronic devices perform medium synchronization of a link according to various embodiments.

(53) FIG. **4**A shows an embodiment in which a first electronic device **401** (e.g., electronic device **310** in FIG. **3**), a second electronic device **402** (e.g., electronic device **310** in FIG. **3**), a third electronic device **403** (e.g., electronic device **310** in FIG. **3**), and/or a fourth electronic device **404** (e.g., electronic device **310** in FIG. **3**) transmit data to an external electronic device **320** using the same link (e.g., second link **332** in FIG. **3**).

(54) According to various embodiments of the disclosure, the first electronic device **401**, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** may transmit data through the CSMA/CA scheme. The first electronic device **401**, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** may perform medium synchronization of the second link **332** before performing data transmission. Medium synchronization of the second link **332** may, for example, refer to a situation in which the state of the second link **332** can be updated in real time. Medium synchronization of the second link **332** may be performed by using a portion of data (e.g., header of data) transmitted through the second link **332**.

(55) The first electronic device **401** supporting CSMA/CA may check whether a specific link is in idle state before transmitting data **411**. The first electronic device **401** may identify whether the second link **332** is in an idle state based on information related to the idle state of the second link **332** included in data transmitted by the external electronic device **320**. The information related to the idle state of the second link **332** may include a CCA status field (clear channel assessment field) and/or a network allocation vector (NAV) configuration field. The information related to the idle state of the second link **332** may be included in a ready to send (RTS) message for requesting data transmission through the second link **332**, or in a clear to send (CTS) message indicating that data transmission is possible through the second link **332**. The first electronic device **401** may identify whether a specific link is in an idle state by referring to the CCA status field and/or the NAV configuration field. The first electronic device **401** may determine whether the second link **332** is physically idle by referring to the CCA status field, and determine whether the second link **332** is logically idle by referring to the NAV configuration field. In response to confirming that a specific link is in an idle state, the first electronic device **401** may activate a timer, and may transmit data **411** to the external electronic device **320** through the second link **332** in response to expiration of the timer after a specified time.

(56) While the first electronic device **401** is transmitting the data **411**, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404**, which support CSMA/CA, may wait without transmitting other data (**412**, **413**, **414**, respectively). The second electronic device **402**, the third electronic device **403** and/or the fourth electronic device **404** may determine the end time of transmission of the data **411** based on a portion of the data **411** transmitted through the second link **332** (e.g., rate field and/or length field included in the PHY header, and duration field included in the MAC header), and may wait a specific time **415** based on the end of transmission of the data **411**. The specific time **415** may means a DIFS (distributed inter frame space).

(57) The second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** may activate timers after the specific time **415** elapses, and may transmit data to the

external electronic device **320** through the second link **332** in response to expiration of the timers. The lengths of the timers set in the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** may be different from each other. According to an embodiment, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** may use randomly set timers to check whether the timers expire and transmit data when the timers expire.

(58) With reference to FIG. **4**A, the second electronic device **402** may activate a timer set to a first period **416**, the third electronic device **403** may activate a timer set to a second period **417**, and the fourth electronic device **404** may activate a timer set to a third period **418**. The first period **416**, the second period **417**, and/or the third period **418** may be different from each other. In FIG. **4**A, the first period **416** may be longer than the third period **418**, and the third period **418** may be longer than the second period **417**.

(59) According to various embodiments of the disclosure, in response to confirming that the third period **417** has expired, the third electronic device **403** may transmit data **419** to the external electronic device **320** through the second link **332**. The second electronic device **402** and/or the fourth electronic device **404** may wait without transmitting other data while the third electronic device **403** is transmitting the data **419**. The second electronic device **402** may store the remaining period **416-*b*** obtained by subtracting the elapsed period **416-*a*** from the first period **416**, and the fourth electronic device **404** may store the remaining period **418-*b*** obtained by subtracting the elapsed period **418-*a*** from the third period **418**. The second electronic device **402** and the fourth electronic device **404** may determine the end time of transmission of the data **419** based on a portion of the data **419** transmitted through the second link **332** (e.g., rate field and/or length field included in the PHY header, and duration field included in the MAC header), and may wait a specific time **420** based on the end of transmission of the data **419**. The specific time **420** may refer to a DIFS (distributed inter frame space).

(60) The second electronic device **402** and/or the fourth electronic device **404** may activate the timers again after the specific time **420** has elapsed. In response to confirming that the remaining period **418-*b*** has expired, the fourth electronic device **404** may transmit the data **423** to the external electronic device **320** through the second link **332**. The second electronic device **402** may sense that the fourth electronic device **404** transmits the data **423** before the remaining period **416-*b*** expires and deactivate the timer again. The second electronic device **420** may store the remaining period **422** obtained by subtracting the elapsed period **421** from the remaining period **416-*b***. The second electronic device **402** may determine the end time of transmission of the data **423** based on a portion of the data **423** transmitted through the second link **332** (e.g., rate field and/or length field included in the PHY header, and duration field included in the MAC header), and may wait a specific time **425** based on the end of transmission of the data **423**. The specific time **425** may means a DIFS (distributed inter frame space).

(61) The second electronic device **402** may activate the timer again after the specific time **425** has elapsed. In response to confirming that the remaining period **422** has expired, the fourth electronic device **404** may transmit the data **424** to the external electronic device **320** through the second link **332**.

(62) In a manner described above, the first electronic device **401**, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** may perform medium synchronization of the second link **332**.

(63) Through the method shown in FIG. **4**A, the first electronic device **401**, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404**, which support CSMA/CA, may transmit data through the second link **332** to the external electronic device **320** without collision. For the CSMA/CA scheme to operate smoothly, the first electronic device **401**, the second electronic device **402**, the third electronic device **403**, and/or the fourth electronic device **404** must always be able to receive data transmitted through the second link **332**.

(64) FIG. **4**B is a diagram illustrating an example in which the electronic device operates in non-simultaneous transmission and reception (non-STR) mode according to various embodiments.
(65) With reference to FIG. **4**B, the wireless LAN system **300** may include an electronic device **310** and/or an external electronic device **320**. According to an embodiment, the electronic device **310** may perform wireless communication with the external electronic device **320** through short-range wireless communication. The wireless communication may refer to various communication methods that both the electronic device **310** and/or the external electronic device **320** can support. For example, the wireless communication may be Wi-Fi. The external electronic device **320** may serve as a base station that provides wireless communication to at least one electronic device **310** located within the communication radius of the wireless LAN system **300**. For example, the external electronic device **320** may include an access point (AP) of IEEE 802.11. The electronic device **310** may include a station (STA) of IEEE 802.11.
(66) According to various embodiments of the disclosure, the electronic device **310** and/or the external electronic device **320** may support multi-link operation (MLO). Multi-link operation may, for example, refer to an operation mode in which data is transmitted or received through a plurality of links (e.g., first link **331** and second link **332**). Multi-link operation is an operation mode to be introduced in IEEE 802.11be, and may be an operation mode in which data is transmitted or received through plural links based on plural bands or channels.
(67) According to various embodiments of the disclosure, the electronic device **310** may include a plurality of communication circuits (e.g., first communication circuit **311** and/or second communication circuit **312**) to support multi-link operation. The first communication circuit **311** may transmit data to the external electronic device **320** through a first link **331** or receive data transmitted by the external electronic device **320** through the first link **331**. The first communication circuit **311** may output or receive a signal of a frequency band corresponding to the first link **331** through a first antenna **313**. The second communication circuit **312** may transmit data to the external electronic device **320** through a second link **332** or receive data transmitted by the external electronic device **320** through the second link **332**. The second communication circuit **312** may output or receive a signal of a frequency band corresponding to the second link **332** through a second antenna **314**.
(68) According to various embodiments of the disclosure, the external electronic device **320** may include a plurality of communication circuits (e.g., third communication circuit **321** and/or fourth communication circuit **322**) to support multi-link operation. The third communication circuit **321** may transmit data to the electronic device **310** through the first link **331** or receive data transmitted by the electronic device **310** through the first link **331**. The third communication circuit **321** may output or receive a signal of a frequency band corresponding to the first link **331** through a third antenna **323**. The fourth communication circuit **322** may transmit data to the electronic device **310** through the second link **332** or receive data transmitted by the electronic device **310** through the second link **332**. The fourth communication circuit **322** may output or receive a signal of a frequency band corresponding to the second link **332** through a fourth antenna **324**.
(69) According to various embodiments of the disclosure, the frequency band of the first link **331** and the frequency band of the second link **333** may be different from each other. For example, the frequency band of the first link **331** may be 2.5 GHz, and the frequency band of the second link **332** may be 5 GHz.
(70) According to various embodiments of the disclosure, the electronic device **310** may fail to sufficiently secure a space **431** between the first antenna **313** and the second antenna **314** due to implementation reasons. According to an embodiment, if the space **431** between the first antenna **313** and the second antenna **314** is not sufficiently secured, a signal output by the first antenna **313** and a signal received by the second antenna **314** may interfere with each other. For example, the second antenna **314** may receive a signal resulting from a combination of the signal received through the second link **332** and a part of the signal output by the first antenna **313**, so that the

quality of the signal received through the second link **332** may be degraded.
(71) According to various embodiments of the disclosure, the electronic device **310** may support non-simultaneous transmission and reception (non-STR) mode to prevent a situation in which the signal output by the first antenna **313** and the signal received by the second antenna **314** interfere with each other. The non-STR mode may refer to a mode in which the electronic device **310** does not receive data through the second link **332** when it transmits data to the external electronic device **320** through the first link **331**. The non-STR mode may support an operation of receiving data through the second link **332** while receiving data through the first link **331** and/or transmitting data through the second link **332** while transmitting data through the first link **331**.
(72) According to various embodiments of the disclosure, when the electronic device **310** operating in non-STR mode transmits data to the external electronic device **320** through the first link **331**, it may not receive data through the second link **332**. As the electronic device **310** fails to receive data through the second link **332**, it may be in a situation where medium synchronization of the second link **332** that may be performed using a portion of data cannot be performed. In this case, the electronic device **310** cannot identify whether another electronic device (e.g., second electronic device **402**, third electronic device **403** and/or fourth electronic device **404**) transmits data through the second link **332**; and, to transmit data through the second link **332**, the electronic device **310** activates the timer again after a specified time (e.g., time out) expires, and transmits data after the timer expires, so that the latency of data transmission may increase.
(73) FIG. **4**C is a diagram illustrating an example in which the electronic device operates in enhanced multi-link with single radio (EMLSR) mode according to various embodiments of the disclosure.
(74) With reference to FIG. **4**C, the wireless LAN system **300** may include an electronic device **310** and/or an external electronic device **320**. According to an embodiment, the electronic device **310** may perform wireless communication with the external electronic device **320** through short-range wireless communication. The wireless communication may refer to various communication methods that both the electronic device **310** and/or the external electronic device **320** can support. For example, the wireless communication may be Wi-Fi. The external electronic device **320** may serve as a base station that provides wireless communication to at least one electronic device **310** located within the communication radius of the wireless LAN system **300**. For example, the external electronic device **320** may include an access point (AP) of IEEE 802.11. The electronic device **310** may include a station (STA) of IEEE 802.11.
(75) According to various embodiments of the disclosure, the electronic device **310** and/or the external electronic device **320** may support multi-link operation (MLO). Multi-link operation may be an operation mode in which data is transmitted or received through a plurality of links (e.g., first link **331** and second link **332**). Multi-link operation is an operation mode to be introduced in IEEE 802.11be, and may be an operation mode in which data is transmitted or received through plural links based on plural bands or channels.
(76) According to various embodiments of the disclosure, the electronic device **310** may include a first communication circuit **311** to support multi-link operation. The first communication circuit **311** may transmit data to the external electronic device **320** through a first link **331** or receive data transmitted by the external electronic device **320** through the first link **331**. The first communication circuit **311** may transmit data to the external electronic device **320** through a second link **332** or receive data transmitted by the external electronic device **320** through the second link **332**. The first communication circuit **311** may output or receive a signal of a frequency band corresponding to the first link **331** through a first antenna **313**, and may output or receive a signal of a frequency band corresponding to the second link **332** through a second antenna **314**.
(77) According to various embodiments of the disclosure, the external electronic device **320** may include a plurality of communication circuits (e.g., third communication circuit **321** and/or fourth communication circuit **322**) to support multi-link operation. The third communication circuit **321**

may transmit data to the electronic device **310** through the first link **331** or receive data transmitted by the electronic device **310** through the first link **331**. The third communication circuit **321** may output or receive a signal of a frequency band corresponding to the first link **331** through a third antenna **323**. The fourth communication circuit **322** may transmit data to the electronic device **310** through the second link **332** or receive data transmitted by the electronic device **310** through the second link **332**. The fourth communication circuit **322** may output or receive a signal of a frequency band corresponding to the second link **332** through a fourth antenna **324**.
(78) According to various embodiments of the disclosure, the frequency band of the first link **331** and the frequency band of the second link **333** may be different from each other. For example, the frequency band of the first link **331** may be 2.5 GHz, and the frequency band of the second link **332** may be 5 GHz.
(79) According to various embodiments of the disclosure, the electronic device **310** may support multi-link operation using plural links through a single communication circuit (e.g., first communication circuit **311**) for implementation reasons. In this case, the electronic device **310** may perform enhanced multi-link single radio (EMLSR) mode in which data having a relatively small size (e.g., control data, RTS frame, CTS frame, ACK message) is transmitted through plural links (e.g., first link **331**, second link **332**), but data having a relatively large size is transmitted through a single link (e.g., first link **331**). The EMLSR mode may refer to a mode in which data having a relatively small size is transmitted and received by using plural links, and data having a relatively large size is transmitted and received by using a single link. When the electronic device **310** operating in EMLSR mode transmits relatively large data to the external electronic device **320** through the first link **331**, it may fail to receive data through the second link **333**.
(80) As the electronic device **310** fails to receive data through the second link **332**, a situation may occur in which medium synchronization of the second link **332** that can be performed using a portion of data cannot be performed. In this case, the electronic device **310** cannot identify whether another electronic device (e.g., second electronic device **402**, third electronic device **403** and/or fourth electronic device **404** in FIG. **4**A) transmits data through the second link **332**; and, to transmit data through the second link **332**, the electronic device **310** activates the timer again after a specified time (e.g., time out) expires, and transmits data after the timer expires, so that the latency of data transmission may increase.
(81) Hereinafter, a description will be given of an embodiment in which an electronic device performs medium synchronization of a second link even when data cannot be received through the second link.
(82) FIG. **5** is a block diagram illustrating an example configuration of an electronic device according to various embodiments of the disclosure.
(83) According to various embodiments of the disclosure, the electronic device (e.g., electronic device **310** in FIG. **3**) may include an antenna **410**, a communication circuit **420** (e.g., first communication circuit **311** or second communication circuit **312** in FIG. **3**), and/or a processor **430** (e.g., processor **120** in FIG. **1**).
(84) According to various embodiments of the disclosure, the antenna **410** may receive a signal transmitted by an external electronic device (e.g., external electronic device **320** in FIG. **3**) or transmit a signal to the external electronic device **320**. When MLO is supported, the antenna **410** may transmit or receive a signal of a frequency band corresponding to the first link (e.g., first link **331** in FIG. **3**), and may transmit or receive a signal of a frequency band corresponding to the second link (e.g., second link **332** in FIG. **3**). The antenna **410** may include plural antennas.
(85) The communication circuit **420** may transmit data to the external electronic device **320** through the first link **331** or receive data transmitted by the external electronic device **320** through the first link **331**. The communication circuit **420** may transmit data to the external electronic device **320** through the second link **332** or receive data transmitted by the external electronic device **320** through the second link **332**. The communication circuit **420** may output or receive a signal of

a frequency band corresponding to the first link **331** through the antenna **410**, and may output or receive a signal of a frequency band corresponding to the second link **332** through the antenna **410**.
(86) The processor **430** may be operably connected to the communication circuit **420** and control the operation of the communication circuit **420**.
(87) The processor **430** may identify whether the first link **331** is in an idle state to transmit data through the first link **331**. The processor **430** may identify whether the first link **331** is in idle state by referring to the clear channel assessment (CCA) field and/or the network allocation vector (NAV) configuration field of data transmitted through the first link **331**. Specifically, the processor **430** may determine whether the first link **331** is physically idle by referring to the CCA status field, and determine whether the first link **331** is logically idle by referring to the NAV configuration field. Upon determining that the first link **331** is in an idle state, the processor **430** may activate a first timer for transmission of the first link **331**. The first timer may be a timer used for medium synchronization of the first link **331**. When the first timer for transmission of the first link **331** expires, the processor **430** may control the communication circuit **420** to transmit data to the external electronic device **320** through the first link **331**.
(88) The processor **430** may identify whether the second link **332** is in an idle state to transmit data through the second link **332**. The processor **430** may determine whether the second link **332** is in an idle state by using information related to the idle state of the second link **332** included in data transmitted through the second link **332**. The information related to the idle state of the second link **332** may include a clear channel assessment (CCA) field and/or a network allocation vector (NAV) configuration field.
(89) The processor **430** may identify whether the second link **332** is in an idle state by referring to the CCA status field and/or the NAV configuration field of data transmitted through the second link **332**. Specifically, the processor **430** may determine whether the second link **332** is physically idle by referring to the CCA status field, and determine whether the second link **332** is logically idle by referring to the NAV configuration field. Upon determining that the second link **332** is in an idle state, the processor **430** may activate a second timer for transmission of the second link **332**. When the second timer for transmission of the second link **332** expires, the processor **430** may control the communication circuit **420** to transmit data to the external electronic device **320** through the second link **332**.
(90) For implementation reasons, the processor **430** may control the communication circuit **420** to transmit data to the external electronic device **320** by using a specific mode among the modes supported by multi-link operation. The specific mode may be one of non-STR mode and EMLSR mode as a mode in which reception of data through the second link **332** is not allowed while data is transmitted to the external electronic device **320** through the first link **331**. The processor **430** may transmit information indicating operating in specific mode to the external electronic device **320** through the first link **331**.
(91) While operating in specific mode, the processor **430** may store information on the second timer used for data transmission through the second link **332** in the memory (e.g., memory **130** in FIG. **1**). The information on the second timer may include the remaining time of the second timer (e.g., similar to remaining time **416**-*b* in FIG. **4**A) for checking whether a specified time has expired. As operating in specific mode, the processor **430** may store information related to the idle state of the second link **332** (e.g., CCA status or NAV configuration) in the memory **130**.
(92) When it is not possible to receive data through the second link **332**, the processor **430** may fail to perform medium synchronization of the second link **332** by using a portion of data received through the second link **332**. To perform medium synchronization of the second link **332**, the processor **430** may refer to a field (e.g., CCA or NAV configuration) included in a portion of data transmitted through the second link **332**. Since the processor **430** cannot receive data through the second link **332**, it cannot refer to the field included in a portion of data transmitted through the second link **332**, so that the medium synchronization of the second link **332** may be in released

state. When the medium synchronization of the second link **332** is in released state, the electronic device **310** may be not able to identify whether the second link **332** is in an idle state or identify the time point at which data can be transmitted through the second link **332**.

(93) According to an embodiment, the processor **430** may receive medium synchronization information of the second link **320** for performing medium synchronization of the second link **332** from the external electronic device **320** through the first link **331**. The medium synchronization information may include information about a timer used for transmission of data through the second link **332**. The information about the second timer may include information for changing the remaining time of the timer and/or information related to the idle state of the second link **332**.

(94) According to an embodiment, the processor **430** may perform (or, maintain) medium synchronization of the second link **332** based on the medium synchronization information received through the first link **331**. Performing medium synchronization of the second link **332** may include changing the remaining time of the second timer used for data transmission through the second link **332**, or setting a restart time of the second timer.

(95) According to an embodiment, the processor **430** may identify a time value indicated by the information for changing the remaining time of the second timer included in the medium synchronization information, and change the remaining time of the second timer by subtracting the time value included in the medium synchronization information from the remaining time of the second timer stored in the memory **130**.

(96) The time value included in the medium synchronization information may, for example, refer to a period during which the second link **332** is in an idle state between the time point when the external electronic device **320** receives data through the first link **331** and the time point when the external electronic device **320** transmits medium synchronization information to the electronic device **310** through the first link **331**. When the second link **332** is in an idle state, the remaining time of the timer used for data transmission through the second link **332** should decrease, but the remaining time of the second timer may be maintained in a state in which data cannot be received through the second link **332**. The processor **430** may change the remaining time of the second timer by subtracting the time value included in the medium synchronization information from the remaining time of the second timer stored in the memory **130**, and may perform (or, maintain) medium synchronization of the second link **332** in a manner of changing the remaining time of the second timer.

(97) The processor **430** may change (or update) the existing information related to the idle state of the second link **332** to information related to the idle state of the second link **332** included in the medium synchronization information. The processor **430** may update the previously stored CCA status information with the CCA status information included in the medium synchronization information received through the first link **331**, and may update the previously stored NAV configuration information with NAV configuration information included in the medium synchronization information received through the first link **331**.

(98) By updating the information related to the idle state of the second link **332**, the processor **430** may determine the start time of the idle state of the second link **332** even in a situation in which data cannot be received through the second link **332**, and may activate the second timer again from the determined time point for the second link **332**.

(99) After data transmission through the first link **331** is completed, the processor **430** may perform medium synchronization of the second link **332** or maintain the medium synchronization state based on the received medium synchronization information. The processor **430** may determine the time point when the second link **332** transitions to an idle state based on the updated information related to the idle state of the second link **332**, and may activate the second timer at the time point of transition to the idle state. Upon confirming that the specified time (or changed time) has expired by using the second timer, the processor **430** may control the communication circuit **420** to transmit data to the external electronic device **320** through the second link **332**.

(100) FIG. **6** is a block diagram illustrating an example configuration of an external electronic device according to various embodiments of the disclosure.
(101) According to various embodiments of the disclosure, the external electronic device (e.g., external electronic device **320** in FIG. **3**) may include a processor **610** (e.g., processor **120** in FIG. **1**), a first communication circuit **621** (e.g., third communication circuit **321** in FIG. **3**), a first antenna **622** (e.g., third antenna **323** in FIG. **3**), a second communication circuit **623** (e.g., fourth communication circuit **322** in FIG. **3**), and a second antenna **624** (e.g., fourth antenna **324** in FIG. **3**).
(102) The external electronic device **320** may include a plurality of communication circuits including the first communication circuit **621** and the second communication circuit **623** to support multi-link operation. The first communication circuit **621** may transmit data to an external electronic device (e.g., electronic device **310** in FIG. **5**) through the first link **331** or may receive data transmitted by the external electronic device **310** through the first link **331**. The first communication circuit **621** may output or receive a signal of a frequency band corresponding to the first link **331** through the first antenna **622**. The second communication circuit **623** may transmit data to the external electronic device **310** through the second link **332** or may receive data transmitted by the external electronic device **310** through the second link **332**. The second communication circuit **623** may output or receive a signal of a frequency band corresponding to the second link **332** through the second antenna **624**.
(103) According to various embodiments of the disclosure, the frequency band of the first link **331** and the frequency band of the second link **333** may be different from each other. For example, the frequency band of the first link **331** may be 2.5 GHz, and the frequency band of the second link **332** may be 5 GHz or 6 GHz.
(104) The processor **610** may be operably connected to the first communication circuit **621** and/or the second communication circuit **623**, and control operations of the first communication circuit **621** and/or the second communication circuit **623**.
(105) The processor **610** may receive data from the electronic device **310** through the first link **331**. The processor **610** may receive information indicating the operation mode of the electronic device **310** from the electronic device **310** during or before receiving data. According to an embodiment, the processor **610** may receive information indicating that the electronic device **310** operates in non-STR mode or EMLSR mode included in messages (e.g., information included in a specific field defined in the multi-link element (e.g., information indicating whether EMLSR is supported included in EML capabilities, NSTR (non-STR) bitmap included in the STA control field) exchanged while establishing the first link **331**.
(106) As the electronic device **310** operates in a specific mode, the processor **610** may generate medium synchronization information for the electronic device **310** to perform medium synchronization of the second link **332**.
(107) The medium synchronization information may include information about a second timer used for transmission of data through the second link **332**. The information about the second timer may include information for changing a remaining time of the second timer and/or information related to the idle state of the second link **332**.
(108) The processor **610** may identify the period during which the second link **332** is in an idle state between the time point when data is received from the electronic device **310** through the first link **331** and the time point when the external electronic device **320** transmits medium synchronization information to the electronic device **310** through the first link **331**. The processor **610** may generate information for changing the remaining time of the second timer including information about the period during which the second link **332** is in idle state (e.g., length of the period or number of slots corresponding to the period).
(109) The processor **610** may identify the CCA status information and the NAV configuration at a time point before transmitting medium synchronization information, and generate information

related to the idle state of the second link **332** including the updated CCA status information and NAV configuration.

(110) The processor **610** may determine whether another external electronic device transmits data at a time point before transmitting medium synchronization information, determine, when the other external electronic device transmits data, the transmission completion time of the data based on a portion of the received data (e.g., PHY header), and update CCA status information. The updated CCA status information may include the remaining time until data transmission is completed, and may indicate that the second link **332** can be transitioned to the idle state after the remaining time expires.

(111) The processor **610** may determine whether another external electronic device transmits data at a time point before transmitting medium synchronization information, determine, when the other external electronic device transmits data, the transmission completion time of the data based on a portion of the received data (e.g., MAC header), and update the NAV configuration. The updated NAV configuration may include the remaining time until data transmission is completed, and may indicate that the second link **332** can be transitioned to the idle state after the remaining time expires.

(112) The processor **610** may control the first communication circuit **621** to transmit the medium synchronization information of the second link **332**, including information for changing a remaining time of the second timer and/or information related to the idle state of the second link **332**, to the electronic device **310** through the first link **331**. The medium synchronization information of the second link **332** may be transmitted in various forms. For example and without limitation, the medium synchronization information of the second link **332** may be transmitted to the electronic device **310** through a message among a response message (ACK) for data received through the first link **331**, a control message associated with the first link **331**, a ready to send (RTS) message requesting data transmission through the first link **331**, and a clear to send (CTS) message indicating that data transmission is possible through the first link **331**.

(113) FIG. **7** is a diagram illustrating an example in which the electronic device performs medium synchronization of the second link by using medium synchronization information received through the first link according to various embodiments of the disclosure.

(114) The electronic device (e.g., electronic device **310** in FIG. **5**) is an electronic device that supports multi-link operation, and may transmit or receive data to or from an external electronic device (e.g., electronic device **600** in FIG. **6**) through plural links including a first link (e.g., first link **331** in FIG. **3**) and a second link (e.g., second link **332** in FIG. **3**).

(115) For implementation reasons, the electronic device **310** may transmit data to the external electronic device **320** by using a specific mode among the modes supported by multi-link operation. The specific mode may be one of non-STR mode and EMLSR mode as a mode in which reception of data through the second link **332** is not allowed while data is transmitted to the external electronic device **320** through the first link **331**. The electronic device **310** may transmit information indicating operation in a specific mode to the external electronic device **320** through the first link **331**.

(116) The electronic device **310** may transmit data **710** to the external electronic device **320** through the first link **331**. The electronic device **310** may be unable to receive data through the second link **332** while operating in the specific mode, in which case the medium synchronization of the second link **332** may be in released state. When the medium synchronization of the second link **332** is in released state, the electronic device **310** may fail to identify whether the second link **332** is in idle state or the time point when data can be transmitted through the second link **332**.

(117) While receiving the data **710**, the external electronic device **320** may generate medium synchronization information for the electronic device **310** to perform medium synchronization of the second link **332**.

(118) The medium synchronization information may include, for example, information about the

second timer used for transmission of data through the second link **332**. The information about the second timer may include for example, information for changing the remaining time of the second timer and/or information related to the idle state of the second link **332**.
(119) The external electronic device **320** may identify the period during which the second link **332** is in idle state between the time point when data is received from the electronic device **310** through the first link **331** and the time point when the external electronic device **320** transmits medium synchronization information to the electronic device **310** through the first link **331**. The external electronic device **320** may generate information for changing the remaining time of the second timer including information about the period during which the second link **332** is in an idle state (e.g., length of the period or number of slots corresponding to the period).
(120) The external electronic device **320** may identify the CCA status information and/or the NAV configuration at a time point before transmitting medium synchronization information, and generate information related to the idle state of the second link **332** including the updated CCA status information and/or NAV configuration.
(121) The external electronic device **320** may determine whether another external electronic device transmits data at a time point before transmitting medium synchronization information, determine, when the other externalelectronic device transmits data, the transmission completion time of the data based on a portion of the received data (e.g., PHY header), and update CCA status information. The other external electronic device may an external electronic device transmitting or receiving the data to the electronic device **310**. The updated CCA status information may include the remaining time until data transmission is completed, and may indicate that the second link **332** can be transitioned to an idle state after the remaining time expires.
(122) The external electronic device **320** may determine whether another external electronic device transmits data at a time point before transmitting medium synchronization information, determine, when the other external electronic device transmits data, the transmission completion time of the data based on a portion of the received data (e.g., MAC header), and update the NAV configuration. The updated NAV configuration may include the remaining time until data transmission is completed, and may indicate that the second link **332** can be transitioned to idle state after the remaining time expires.
(123) The external electronic device **320** may transmit the medium synchronization information of the second link **332**, including information for changing the remaining time of the second timer and/or information related to the idle state of the second link **332**, to the electronic device **310** through the first link **331**. Although the medium synchronization information of the second link **332** is shown in FIG. **7** as being included in a response message **720** indicating reception of the data **710**, the medium synchronization information of the second link **332** may be transmitted in various forms. For example and without limitation, the medium synchronization information of the second link **332** may be transmitted to the electronic device **310** through a message among a response message **720** for data received through the first link **331**, a control message associated with the first link **331**, a ready to send (RTS) message requesting data transmission through the first link **331**, and a clear to send (CTS) message indicating that data transmission is possible through the first link **331**.
(124) The electronic device **310** may receive medium synchronization information of the second link **332** through the response message **720** and perform (or maintain) medium synchronization of the second link **332**.
(125) The electronic device **310** may change the remaining time of the second timer by subtracting the time value included in the medium synchronization information from the remaining time of the second timer, and may perform (or maintain) medium synchronization of the second link **332** by changing the remaining time of the second timer.
(126) The electronic device **310** may change (or update) the existing information related to the idle state of the second link **332** to information related to the idle state of the second link **332** included

in the medium synchronization information. The electronic device **310** may update the previously stored CCA status information with the CCA status information included in the medium synchronization information received through the first link **331**, and may update the previously stored NAV configuration information with NAV configuration information included in the medium synchronization information received through the first link **331**.

(127) By updating the information related to the idle state of the second link **332**, the electronic device **310** may determine the start time of the idle state of the second link **332** even in a situation where data cannot be received through the second link **332**, and may activate the timer again from the determined time point for the second link **332**.

(128) FIG. **8** is a diagram illustrating an example structure of a frame including information for the electronic device to perform medium synchronization of a second link according to various embodiments of the disclosure.

(129) A frame **800** including information for performing medium synchronization of the second link may include: a header **810** including various fields such as a field related to an encoding scheme (e.g., orthogonal frequency-division multiplexing, OFDMA) of data through the second link **332** and a field indicating the length of the frame **800**; a packet service data unit (PSDU) **820** including data to be transmitted; and medium synchronization information **830**.

(130) The medium synchronization information **830** may include a field **831** indicating the idle period of the second link **332**, a CCA status field **832**, and/or a NAV configuration field **833**.

(131) The field **831** indicating the idle period of the second link **332** may include the period during which the second link **332** is in idle state between the time point when the external electronic device **320** receives data through the first link **331** and the time point when the external electronic device **320** transmits medium synchronization information to the electronic device **310** through the first link **331**.

(132) The CCA status field **832** may include a transmission completion time of data in a situation where another external electronic device transmits data through the second link **332**.

(133) The NAV configuration field **833** is a field including information for determining whether the second link **332** is logically in idle state, and may include a time point at which whether the second link **332** is in idle state can be determined.

(134) An electronic device (e.g., electronic device **310** in FIG. **5**) according to various embodiments of the disclosure may include: at least one antenna (e.g., antenna **410** in FIG. **5**); a communication circuit (e.g., communication circuit **420** in FIG. **5**) electrically connected to the antenna **410** and configured to transmit and receive data through a first link (e.g., first link **331** in FIG. **4**B) and/or second link (e.g., second link **332** in FIG. **4**B) established between an external electronic device (e.g., external electronic device **320** in FIG. **4**B) and the electronic device **310**; and a processor (e.g., processor **430** in FIG. **5**) operably connected to the communication circuit **420**, wherein the processor **430** may be configured to: receive medium synchronization information of the second link **332** through the first link **331** while transmitting data to the external electronic device **320** through the first link **331**; and perform synchronization of the second link **332** based on the medium synchronization information.

(135) In the electronic device **310** according to various embodiments of the disclosure, the processor **430** may be configured to control the communication circuit **420** to receive the medium synchronization information through the first link **331** when operating in an operation mode in which data reception through the second link **332** is not allowed while data is transmitted through the first link **331**.

(136) In the electronic device **310** according to various embodiments of the disclosure, the processor **430** may be configured to control the communication circuit **420** to transmit information about the operation mode to the external electronic device **320**.

(137) In the electronic device **310** according to various embodiments of the disclosure, the medium synchronization information may include information about a timer for medium synchronization of

the second link **332**.

(138) In the electronic device **310** according to various embodiments of the disclosure, the processor **430** may be configured to perform medium synchronization of the second link **332** by changing the remaining time of the timer or setting a restart time of the timer based on the information about the timer.

(139) In the electronic device **310** according to various embodiments of the disclosure, the information about the timer may include information for changing the remaining time of the timer and information for determining a restart time of the timer.

(140) In the electronic device **310** according to various embodiments of the disclosure, the information for changing the remaining time of the timer may include the period during which the second link **332** is in an idle state between the time point when data is transmitted to the external electronic device **320** through the first link **331** and the time point when the medium synchronization information is transmitted to the electronic device **310**.

(141) In the electronic device **310** according to various embodiments of the disclosure, the processor **430** may be configured to perform medium synchronization of the second link **332** by subtracting the period from the remaining time of the counter.

(142) In the electronic device **310** according to various embodiments of the disclosure, the processor **430** may be configured to determine the restart time of the counter based on the information for determining the restart time of the timer.

(143) An electronic device (e.g., external electronic device **320** in FIG. **4**B) according to various embodiments of the disclosure may include: at least one antenna (e.g., first antenna **622**, second antenna **624** in FIG. **6**); a communication circuit (e.g., first communication circuit **621** and/or second communication circuit **623** in FIG. **6**) electrically connected to the antenna **622** or **624** and configured to transmit and receive data through a first link (e.g., first link **331** in FIG. **4**B) and/or second link (e.g., second link **332** in FIG. **4**B) established between an external electronic device (e.g., electronic device **310** in FIG. **4**B) and the electronic device **320**; and a processor (e.g., processor **610** in FIG. **6**) operably connected to the communication circuit **621** or **623**, wherein the processor **610** may be configured to: generate medium synchronization information of the second link **332** while receiving data from the external electronic device **310** through the first link **331**; and control, upon completion of reception of the data, the communication circuit **621** or **623** to transmit the medium synchronization information of the second link **332** to the external electronic device **310** through the first link **331**, and wherein the medium synchronization information may include information about a timer for medium synchronization of the second link **332**.

(144) In the electronic device **320** according to various embodiments of the disclosure, the processor **610** may be configured to: receive information indicating that the external electronic device **320** operates in an operation mode where data reception through the second link **332** is not allowed while transmitting data to the electronic device **320** through the first link **331**; and transmit, in response to receiving the information, medium synchronization information of the second link **332** through the first link **331**.

(145) In the electronic device **320** according to various embodiments of the disclosure, the information about the timer may include information for changing the remaining time of the timer and information for determining a restart time of the timer.

(146) In the electronic device **320** according to various embodiments of the disclosure, the processor **610** may be configured to: identify the period during which the second link **332** is in idle state between the time point when data is transmitted to the external electronic device **310** through the first link **331** and the time point when the medium synchronization information is transmitted to the electronic device **320**; and generate information for changing the remaining time of the timer based on the identified period.

(147) In the electronic device **320** according to various embodiments of the disclosure, the processor **610** may be configured to generate information for determining the restart time of the

timer based on the length of a frame transmitted through the second link **332** to another external electronic device.

(148) FIG. **9** is a flowchart illustrating an example method of operating an electronic device according to various embodiments of the disclosure.

(149) At operation **910**, the electronic device (e.g., electronic device **310** in FIG. **5**) may transmit data to an external electronic device (e.g., electronic device **320** in FIG. **6**) through a first link (e.g., first link **331** in FIG. **3**).

(150) The electronic device (e.g., electronic device **310** in FIG. **5**) is an electronic device that supports multi-link operation, and may transmit or receive data to or from an external electronic device (e.g., electronic device **320** in FIG. **6**) through plural links including a first link (e.g., first link **331** in FIG. **3**) and a second link (e.g., second link **332** in FIG. **3**).

(151) For implementation reasons, the electronic device **310** may transmit data to the external electronic device **320** by using a specific mode among the modes supported by multi-link operation. The specific mode may be one of non-STR mode and EMLSR mode as a mode in which reception of data through the second link **332** is not allowed while data is transmitted to the external electronic device **320** through the first link **331**. The electronic device **310** may transmit information indicating operation in specific mode to the external electronic device **320** through the first link **331**.

(152) The electronic device **310** may be unable to receive data through the second link **332** while operating in the specific mode, in which case the medium synchronization of the second link **332** may be in released state. When the medium synchronization of the second link **332** is in released state, the electronic device **310** may fail to identify whether the second link **332** is in an idle state or the time point when data can be transmitted through the second link **332**.

(153) At operation **920**, the electronic device **310** may store information about a second timer for synchronization of the second link (e.g., second link **332** in FIG. **3**) in the memory (e.g., memory **130** in FIG. **1**).

(154) While operating in specific mode, the electronic device **310** may store information on the second timer used for data transmission through the second link **332** in the memory (e.g., memory **130** in FIG. **1**). The information on the second timer may include the remaining time of the second timer (e.g., like the remaining time **416-*b*** in FIG. **4**A) for checking whether a specified time has expired. While operating in specific mode, the processor **430** may store information related to the idle state of the second link **332** (e.g., CCA status or NAV configuration) in the memory **130**.

(155) At operation **930**, the electronic device **310** may receive medium synchronization information of the second link **332** through the first link **331**.

(156) The medium synchronization information may include information about the second timer used for transmission of data through the second link **332**. The information about the second timer may include information for changing the remaining time of the second timer and/or information related to the idle state of the second link **332**.

(157) The external electronic device **320** may transmit the medium synchronization information of the second link **332**, including information for changing the remaining time of the second timer and/or information related to the idle state of the second link **332**, to the electronic device **310** through the first link **331**.

(158) The medium synchronization information of the second link **332** may be transmitted in various forms. For example and without limitation, the medium synchronization information of the second link **332** may be transmitted to the electronic device **310** through a message among a response message (ACK) for data received through the first link **331**, a control message associated with the first link **331**, a ready to send (RTS) message requesting data transmission through the first link **331**, and a clear to send (CTS) message indicating that data transmission is possible through the first link **331**.

(159) At operation **940**, the electronic device **310** may perform medium synchronization of the

second link based on the medium synchronization information.
(160) The electronic device **310** may change the remaining time of the second timer by subtracting the time value included in the medium synchronization information from the remaining time of the second timer, and may perform (or maintain) medium synchronization of the second link **332** by changing the remaining time of the second timer.
(161) The electronic device **310** may change (or update) the existing information related to the idle state of the second link **332** to information related to the idle state of the second link **332** included in the medium synchronization information. The electronic device **310** may update the previously stored CCA status information with the CCA status information included in the medium synchronization information received through the first link **331**, and may update the previously stored NAV configuration information with NAV configuration information included in the medium synchronization information received through the first link **331**.
(162) By updating the information related to the idle state of the second link **332**, the electronic device **310** may determine the start time of the idle state of the second link **332** even in a situation where data cannot be received through the second link **332**, and may activate the timer again from the determined time point for the second link **332**.
(163) FIG. **10** is a flowchart illustrating an example method of operating an electronic device according to various embodiments of the disclosure.
(164) At operation **1010**, the electronic device (e.g., external electronic device **320** in FIG. **6**) may receive data from an external electronic device (e.g., electronic device **310** in FIG. **5**) through the first link (e.g., first link **331** in FIG. **3**).
(165) At operation **1020**, the electronic device **320** may generate medium synchronization information of the second link (e.g., second link **332** in FIG. **3**).
(166) The medium synchronization information may include information about the second timer used for transmission of data through the second link **332**. The information about the second timer may include information for changing the remaining time of the second timer and/or information related to the idle state of the second link **332**.
(167) The electronic device **320** may identify the period during which the second link **332** is in idle state between the time point when data is received from the external electronic device **310** through the first link **331** and the time point when the electronic device **320** transmits medium synchronization information to the external electronic device **310** through the first link **331**. The electronic device **320** may generate information for changing the remaining time of the second timer including information about the period during which the second link **332** is in an idle state (e.g., length of the period or number of slots corresponding to the period).
(168) The electronic device **320** may identify the CCA status information and the NAV configuration at a time point before transmitting medium synchronization information, and generate information related to the idle state of the second link **332** including the updated CCA status information and NAV configuration.
(169) The electronic device **320** may determine whether another external electronic device transmits data at a time point before transmitting medium synchronization information, determine, when the other external electronic device transmits data, the transmission completion time of the data based on a portion of the received data (e.g., PHY header), and update CCA status information. The updated CCA status information may include the remaining time until data transmission is completed, and may indicate that the second link **332** can be transitioned to idle state after the remaining time expires.
(170) The electronic device **320** may determine whether another external electronic device transmits data at a time point before transmitting medium synchronization information, determine, when the other external electronic device transmits data, the transmission completion time of the data based on a portion of the received data (e.g., MAC header), and update the NAV configuration. The updated NAV configuration may include the remaining time until data transmission is

completed, and may indicate that the second link **332** can be transitioned to idle state after the remaining time expires.
(171) At operation **1030**, the electronic device **320** may transmit the medium synchronization information to the external electronic device **310** through the first link **331**.
(172) The electronic device **320** may transmit the medium synchronization information of the second link **332**, including information for changing the remaining time of the second timer and/or information related to the idle state of the second link **332**, to the external electronic device **310** through the first link **331**. The medium synchronization information of the second link **332** may be transmitted in various forms. For example and without limitation, the medium synchronization information of the second link **332** may be transmitted to the electronic device **310** through a message among a response message **720** for data received through the first link **331**, a control message associated with the first link **331**, a ready to send (RTS) message requesting data transmission through the first link **331**, and a clear to send (CTS) message indicating that data transmission is possible through the first link **331**.
(173) An operation method of an electronic device (e.g., electronic device **310** in FIG. **5**) according to various embodiments of the disclosure may include: receiving, in a state of transmitting data through a first link (e.g., first link **331** in FIG. **4**B) established between an external electronic device (e.g., external electronic device **320** in FIG. **4**B) and the electronic device **310**, medium synchronization information of a second link (e.g., second link **332** in FIG. **4**B); and performing synchronization of the second link **332** based on the medium synchronization information.
(174) In the operation method of the electronic device **310** according to various embodiments of the disclosure, receiving medium synchronization information of a second link **332** may include receiving the medium synchronization information through the first link **331** when operating in an operation mode in which data reception through the second link **332** is not allowed while data is transmitted through the first link **331**.
(175) In the operation method of the electronic device **310** according to various embodiments of the disclosure, the medium synchronization information may include information about a timer for medium synchronization of the second link **332**.
(176) In the operation method of the electronic device **310** according to various embodiments of the disclosure, the information about the timer may include information for changing the remaining time of the timer and information for determining a restart time of the timer.
(177) In the operation method of the electronic device **310** according to various embodiments of the disclosure, the information for changing the remaining time of the timer may include the period during which the second link **332** is in idle state between the time point when data is transmitted to the external electronic device **320** through the first link **331** and the time point when the medium synchronization information is transmitted to the electronic device **310**, and performing synchronization of the second link **332** may include performing medium synchronization of the second link **332** by subtracting the period from the remaining time of the counter.
(178) In the operation method of the electronic device **310** according to various embodiments of the disclosure, performing synchronization of the second link **332** may include determining the restart time of the counter based on the information for determining the restart time of the timer.
(179) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance, or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.
(180) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be

used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as "A or B," "at least one of A and B," "at least one of A or B," "A, B, or C," "at least one of A, B, and C," and "at least one of A, B, or C," may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as "1st" and "2nd," or "first" and "second" may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term "operatively" or "communicatively", as "coupled with," "coupled to," "connected with," or "connected to" another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.
(181) As used in connection with various embodiments of the disclosure, the term "module" may include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, "logic," "logic block," "part," or "circuitry". A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).
(182) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the "non-transitory" storage medium is a tangible device, and may not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.
(183) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
(184) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be

carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.
(185) While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device comprising: at least one antenna; a communication circuit electrically connected to the at least one antenna; and a processor operably connected to the communication circuit, wherein the at least one processor is configured, individually or collectively, to: establish via the communication circuit a multi-link operation in which data is transmitted and/or received through plural data links between the electronic device and an external electronic device, the plural data links including a first link and a second link for transmitting and receiving data between the electronic device and the external electronic device; and in an operation mode of the electronic device, during the multi-link operation, in which data reception through the second link is not allowed while data is transmitted through the first link: receive, in a state of transmitting data to the external electronic device through the first link, medium synchronization information of the second link through the first link, wherein the medium synchronization information includes timer information about a timer for medium synchronization of the second link, wherein the timer information includes first information for changing a remaining time of the timer; and perform synchronization of the second link based on the medium synchronization information.
2. The electronic device of claim 1, wherein the at least one processor is configured, individually or collectively, to control the communication circuit to transmit information about the operation mode to the external electronic device.
3. The electronic device of claim 1, wherein the at least one processor is configured, individually or collectively, to perform medium synchronization of the second link by changing a remaining time of the timer or setting a restart time of the timer, based on the timer information.
4. The electronic device of claim 1, wherein the timer information further includes second information for determining a restart time of the timer.
5. The electronic device of claim 4, wherein the first information includes a period during which the second link is in an idle state between a time point when data is transmitted to the external electronic device through the first link and a time point when the medium synchronization information is transmitted to the electronic device.
6. The electronic device of claim 5, wherein the at least one processor is configured, individually or collectively, to perform medium synchronization of the second link by subtracting the period from the remaining time of the timer.
7. The electronic device of claim 4, wherein the at least one processor is configured, individually or collectively, to determine the restart time of the timer based on the second information.
8. A method of operating an electronic device, the method comprising: establishing a multi-link operation in which data is transmitted and/or received through plural data links between the electronic device and an external electronic device, the plural data links including a first link and a second link for transmitting and receiving data between the electronic device and the external electronic device; and in an operation mode of the electronic device, during the multi-link operation, in which data reception through the second link is not allowed while data is transmitted through the first link: receiving, in a state of transmitting data through a first link established

between an external electronic device and the electronic device, medium synchronization information of a second link through the first link, wherein the medium synchronization information includes timer information about a timer for medium synchronization of the second link, wherein the timer information includes first information for changing a remaining time of the timer; and performing synchronization of the second link based on the medium synchronization information.

9. The method of claim 8, wherein the timer information further includes second information for determining a restart time of the timer.

10. The method of claim 9, wherein: the first information includes a period during which the second link is in an idle state between a time point when data is transmitted to the external electronic device through the first link and a time point when the medium synchronization information is transmitted to the electronic device; and the method comprises performing medium synchronization of the second link by subtracting the period from the remaining time of the timer.

11. The method of claim 9, comprising determining the restart time of the timer based on the second information.

12. An electronic device comprising: at least one antenna; a communication circuit electrically connected to the at least one antenna; and at least one processor operably connected to the communication circuit, wherein the at least one processor is configured, individually or collectively, to: establish via the communication circuit a multi-link operation in which data is transmitted and/or received through plural data links between the electronic device and an external electronic device, the plural data links including a first link and a second link; and in an operation mode of the external electronic device, during the multi-link operation, in which data reception through the second link is not allowed while data is transmitted through the first link: generate, in a state of receiving data from the external electronic device through the first link, medium synchronization information of the second link, wherein the medium synchronization information includes timer information about a timer for medium synchronization of the second link, wherein the timer information includes first information for changing a remaining time of the timer; and control, upon completion of receiving of the data, the communication circuit to transmit the medium synchronization information of the second link to the external electronic device through the first link.

13. The electronic device of claim 12, wherein the at least one processor is configured, individually or collectively, to: receive information indicating that the external electronic device operates in the operation mode in which data reception through the second link is not allowed while data is transmitted through the first link; and transmit, in response to receiving the information, the medium synchronization information of the second link through the first link.

14. The electronic device of claim 12, wherein the timer information further includes second information for determining a restart time of the timer.

15. The electronic device of claim 14, wherein the at least one processor is configured, individually or collectively, to: identify a period during which the second link is in an idle state between a time point when data is transmitted to the external electronic device through the first link and a time point when the medium synchronization information is transmitted to the electronic device; and generate the first information based on the identified period.

16. The electronic device of claim 14, wherein the at least one processor is configured, individually or collectively, to generate information for determining the restart time of the timer based on a length of a frame transmitted through the second link to another external electronic device.